



PSYCH 201B

Statistical Intuitions for Social Scientists

Linear Algebra & GLM

Today's Plan

1. Final Project Overview
2. Essence of GLM Linear Algebra
3. Multiple Predictors

Final project details on course website!

Linear Algebra Essentials

**There's an interactive explorer
on the website!**

Linear Algebra Essentials

- Two interpretations of a **Matrix**
 - A table of numbers storing data, coefficients, etc in compact form
 - A **machine** that transforms vectors into other vectors capturing the essence of the linear transformations

For example:

$$A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 4 \\ 5 \end{bmatrix}.$$

Multiplying:

$$A\mathbf{x} = \begin{bmatrix} 2 \times 4 + 0 \times 5 \\ 1 \times 4 + 3 \times 5 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \end{bmatrix}.$$

Here, the matrix is acting as a machine that takes input (4, 5) and outputs (8, 19). The “machine rules” are encoded in the rows of A.

Linear Algebra Essentials

- Two interpretations of a Matrix
 - A table of numbers storing data, coefficients, etc in compact form
 - A **machine** that transforms vectors into other vectors capturing the essence of the linear transformations
- This is the key to linear algebra!
 - A dataset stored as a table can be interpreted as a transformation
 - A transformation once understood can be encoded as a table

Linear Algebra Essentials

- What does the machine do?
 - Each entry of the output is the dot-product (similarity) between one *row* of A and the vector $\mathbf{x}$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

$$A\mathbf{x} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}.$$

Linear Algebra Essentials

- What does the machine do?
 - **Math:** $A\mathbf{x}$ is the linear combination of columns of A

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

$$A = \begin{bmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix},$$

$$A\mathbf{x} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{bmatrix}.$$

$$A\mathbf{x} = x_1a_1 + x_2a_2 + \cdots + x_na_n.$$

Linear Algebra Essentials

- What does the machine do?
 - **Geometry:** it transforms x
 - Scaling: magnitude (of predictor)
 - Reflecting: sign (of predictor)
 - Rotating: mixing (predictors)
- **Simple example:**
 - Every 2x2 or 3x3 matrix = some transformation in a 2d plane or 3d space!
 - Scaling: $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ doubles lengths.
 - Reflection: $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ flips across the x-axis.
 - Rotation: $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ rotates vectors by θ .

Recap: Design Matrix

Dataset: Credit-Card Debt

index	income	limit	rating	cards	age	education	gender	student	married	ethnicity	balance
1	14.89	3606	283	2	34	11	Male	No	Yes	Caucasian	333
2	106.03	6645	483	3	82	15	Female	Yes	Yes	Asian	903
3	104.59	7075	514	4	71	11	Male	No	No	Asian	580
4	148.92	9504	681	3	36	11	Female	No	No	Asian	964
5	55.88	4897	357	2	68	16	Male	No	Yes	Caucasian	331
6	80.18	8047	569	4	77	10	Male	No	No	Caucasian	1151
7	21.00	3388	259	2	37	12	Female	No	No	African American	203
8	71.41	7114	512	2	87	9	Male	No	No	Asian	872
9	15.12	3300	266	5	66	13	Female	No	No	Caucasian	279
10	71.06	6819	491	3	41	19	Female	Yes	Yes	African American	1350

variable	description
income	in thousand dollars
limit	credit limit
rating	credit rating
cards	number of credit cards
age	in years
education	years of education
gender	male or female
student	student or not
married	married or not
ethnicity	African American, Asian, Caucasian
balance	average credit card debt in dollars

But this isn't a number...



Do **students** have a different **credit card balance** from **non-students**?

How does the GLM **see** different variables?



Continuous Variables



Categorical Variables

**Many
simultaneous
continuous
variables**



**Many
simultaneous
categorical
variables**

Design Matrix: The “eyes” of the GLM

$$y = X\beta + \epsilon$$

- **Categorical Variables**
 - We can **encode** them into numbers using a variety of approaches
 - A variable with **k unique levels** is represented with **k-1 columns** in the Design Matrix

Design Matrix: The “eyes” of the GLM

- Categorical Variables
 - We can **encode** them into numbers using a variety of approaches
 - A variable with **k unique levels** is represented with **k-1 columns** in the Design Matrix

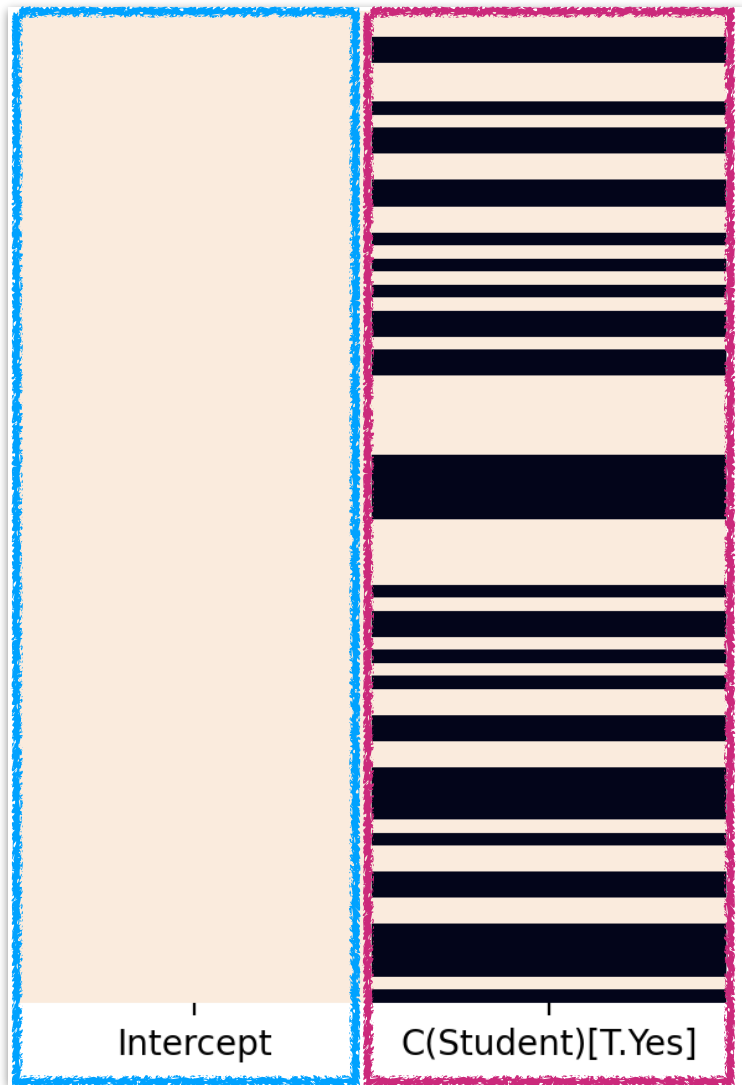
Balance	Student
i64	str
16	"Yes"
1216	"Yes"
148	"No"
108	"No"
532	"Yes"

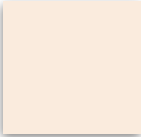
Yes = 1

No = 0

```
array([[1., 1.],  
       [1., 1.],  
       [1., 0.],  
       [1., 0.],  
       [1., 1.]])
```

Design Matrix: The “eyes” of the GLM



 = 1

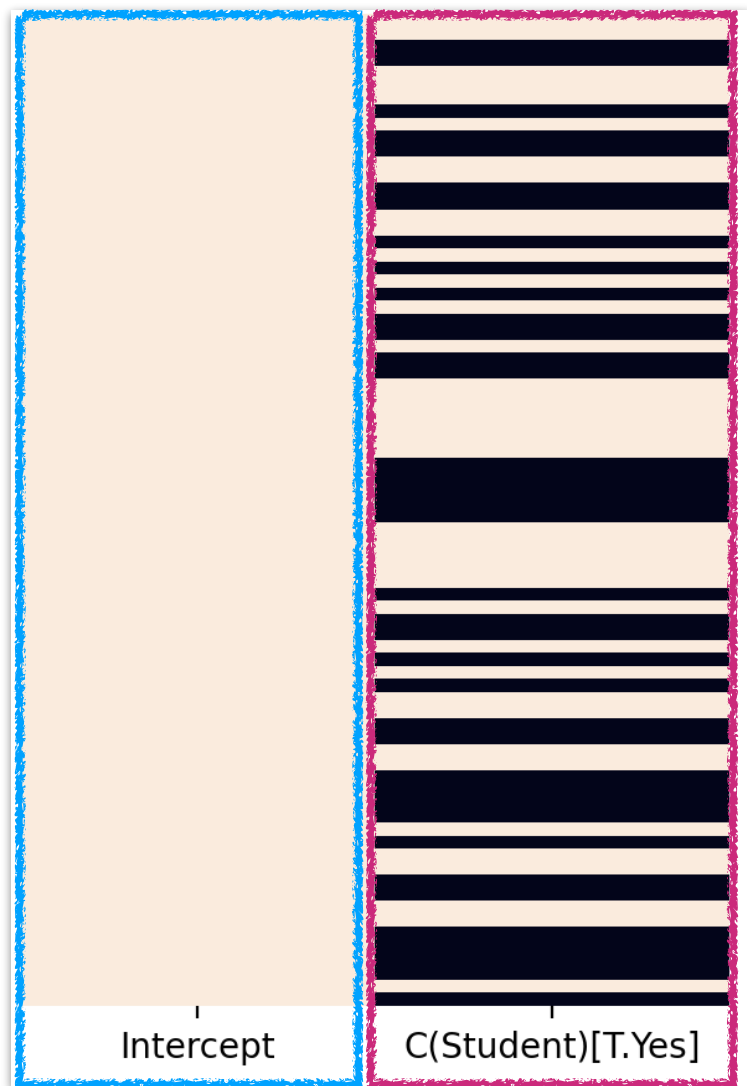
 = 0

(mean of No)(Yes - No)

Balance	Student
i64	str
16	"Yes"
1216	"Yes"
148	"No"
108	"No"
532	"Yes"

```
[[1., 1.],  
 [1., 1.],  
 [1., 0.],  
 [1., 0.],  
 [1., 1.]]
```

Encoding 1: Treatment (Dummy) Codes



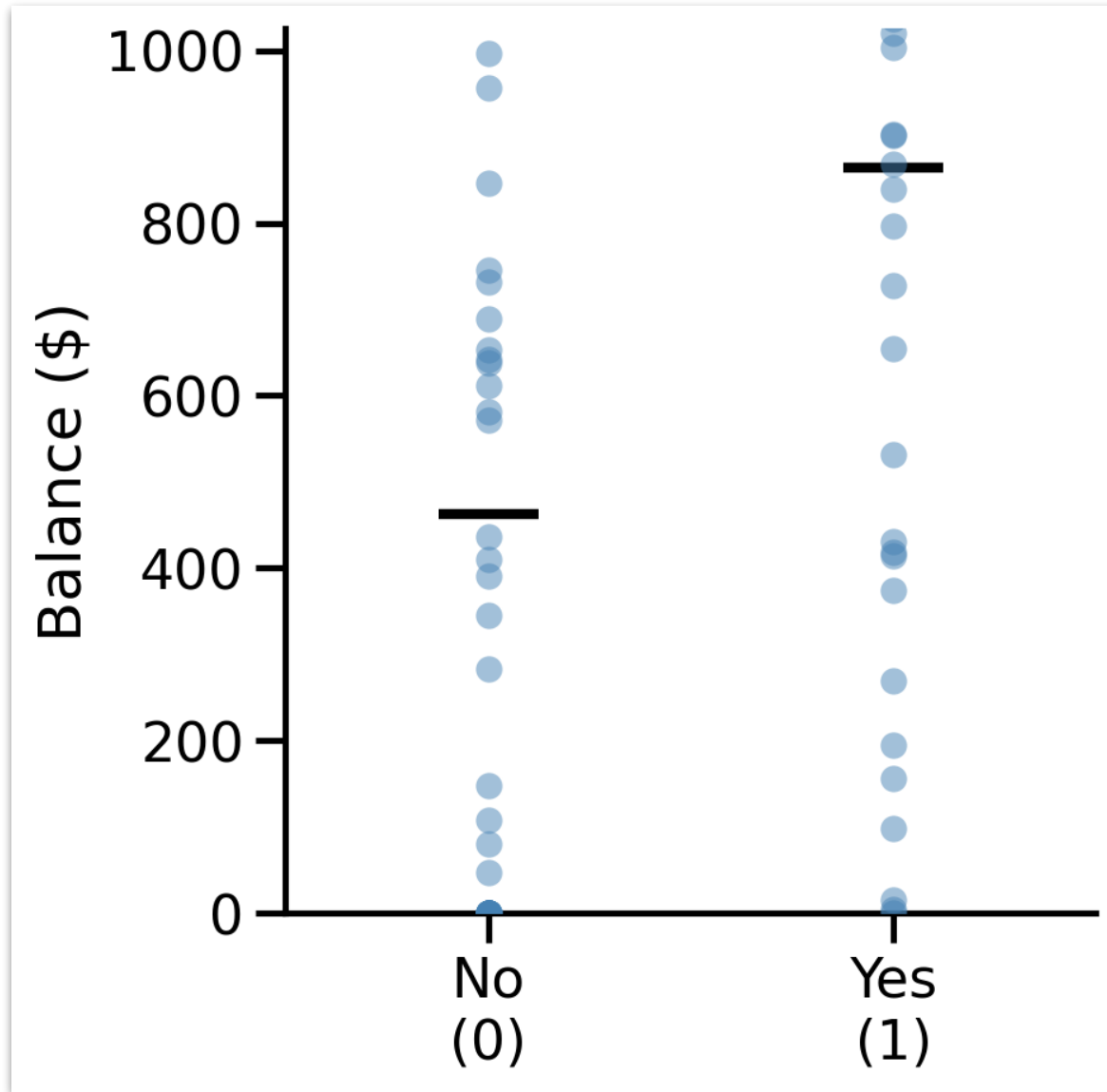
(mean of No)(Yes - No)

Balance	Student
i64	str
16	"Yes"
1216	"Yes"
148	"No"
108	"No"
532	"Yes"

```
[[1., 1.],
 [1., 1.],
 [1., 0.],
 [1., 0.],
 [1., 1.]]
```

- **Reference level** is coded as 0, other level(s) coded as 1
- **Intercept** = reference-level (mean)
- **Slope(s)** = difference from reference-level mean

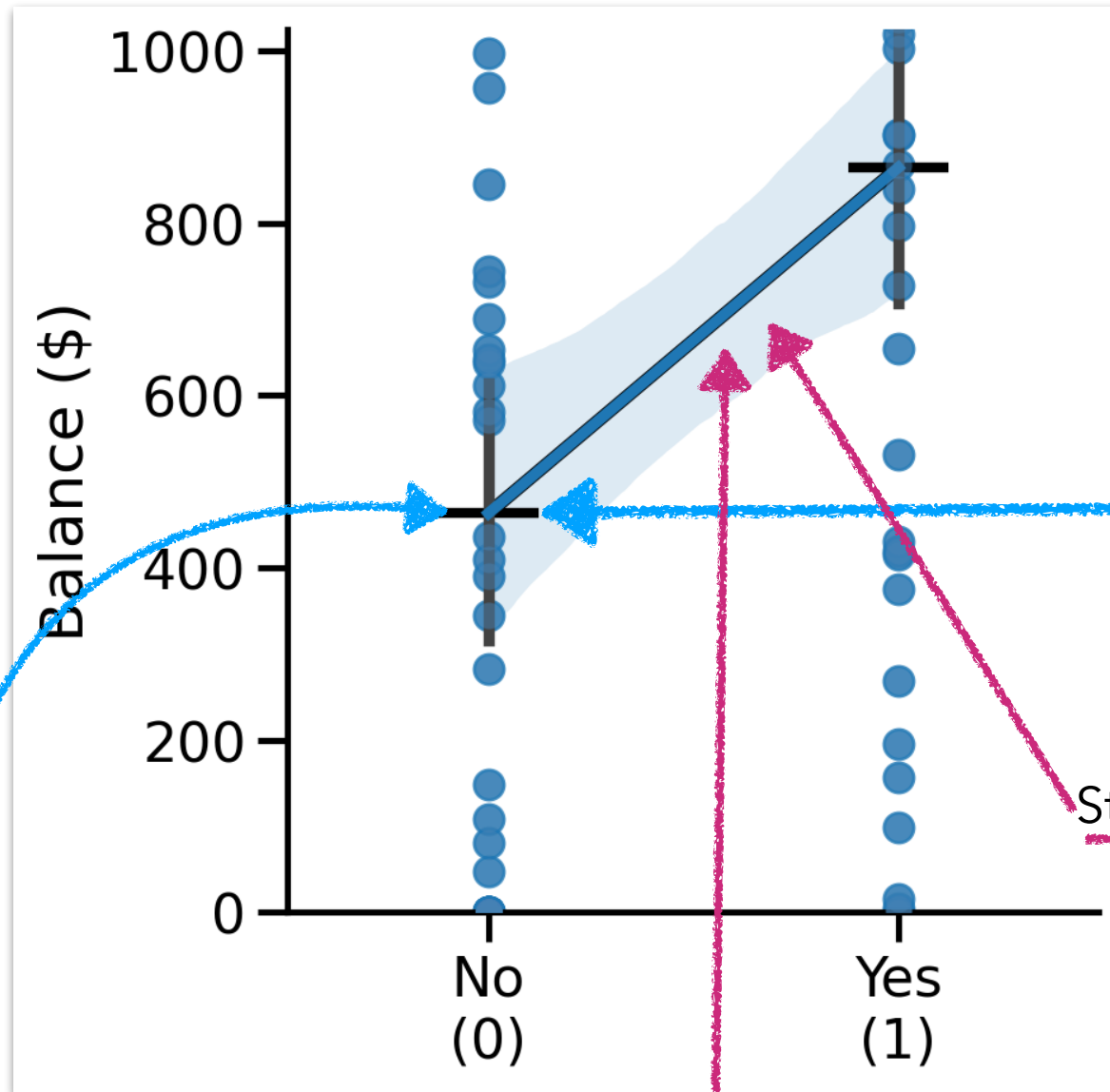
Treatment (Dummy) Coding Visually



0: Student = No
1: Student = Yes

- **Reference level** is coded as 0, other level(s) coded as 1
- **Intercept** = reference-level (mean)
- **Slope(s)** = difference from reference-level mean

Treatment (Dummy) Coding Visually

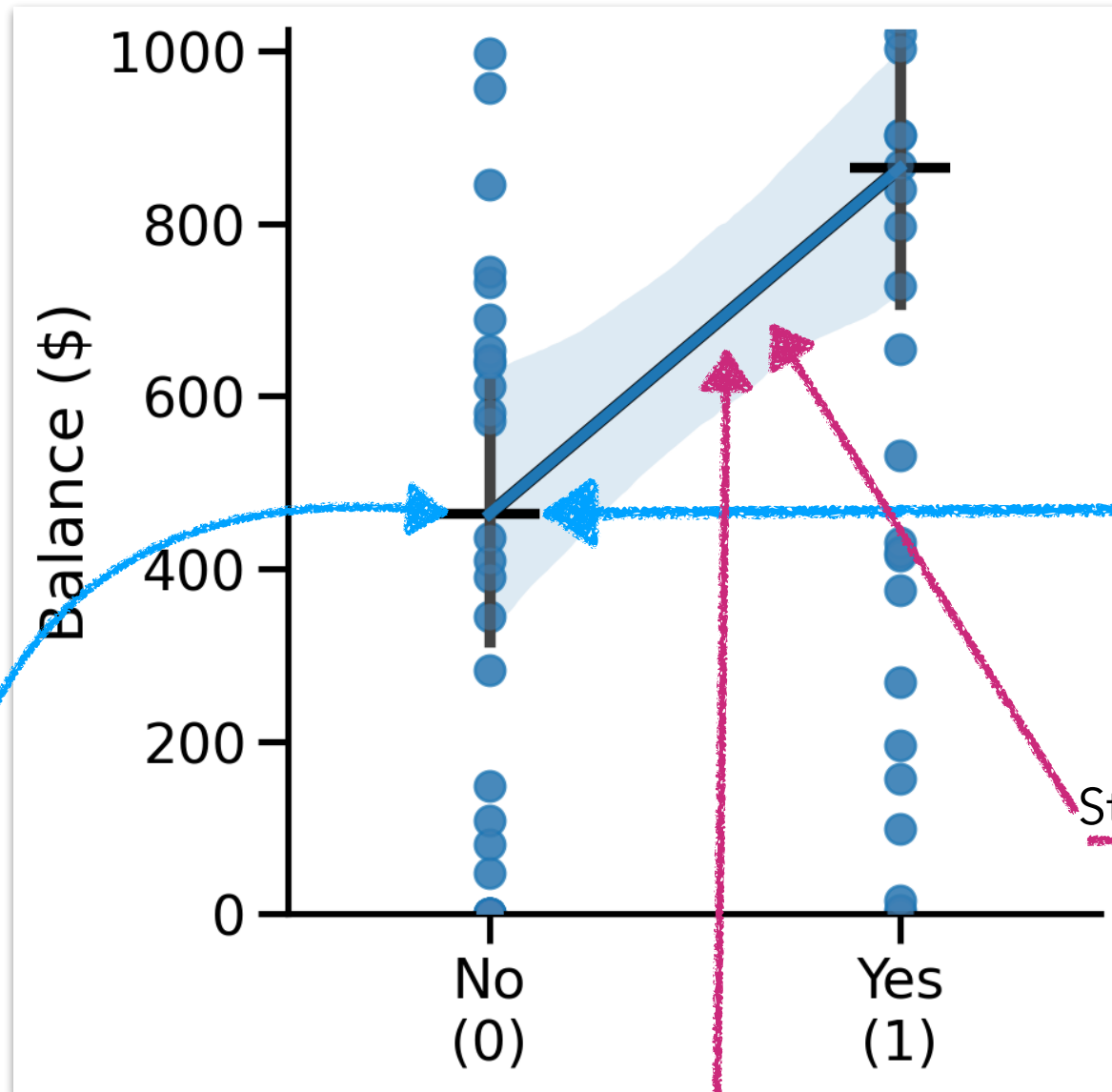


0: Student = No
1: Student = Yes

Student (Yes) - Student (No)

- **Reference level** is coded as 0, other level(s) coded as 1
- Intercept = reference-level (mean)
- **Slope(s)** = difference from reference-level mean

Linear Model (GLM) of Mean Differences

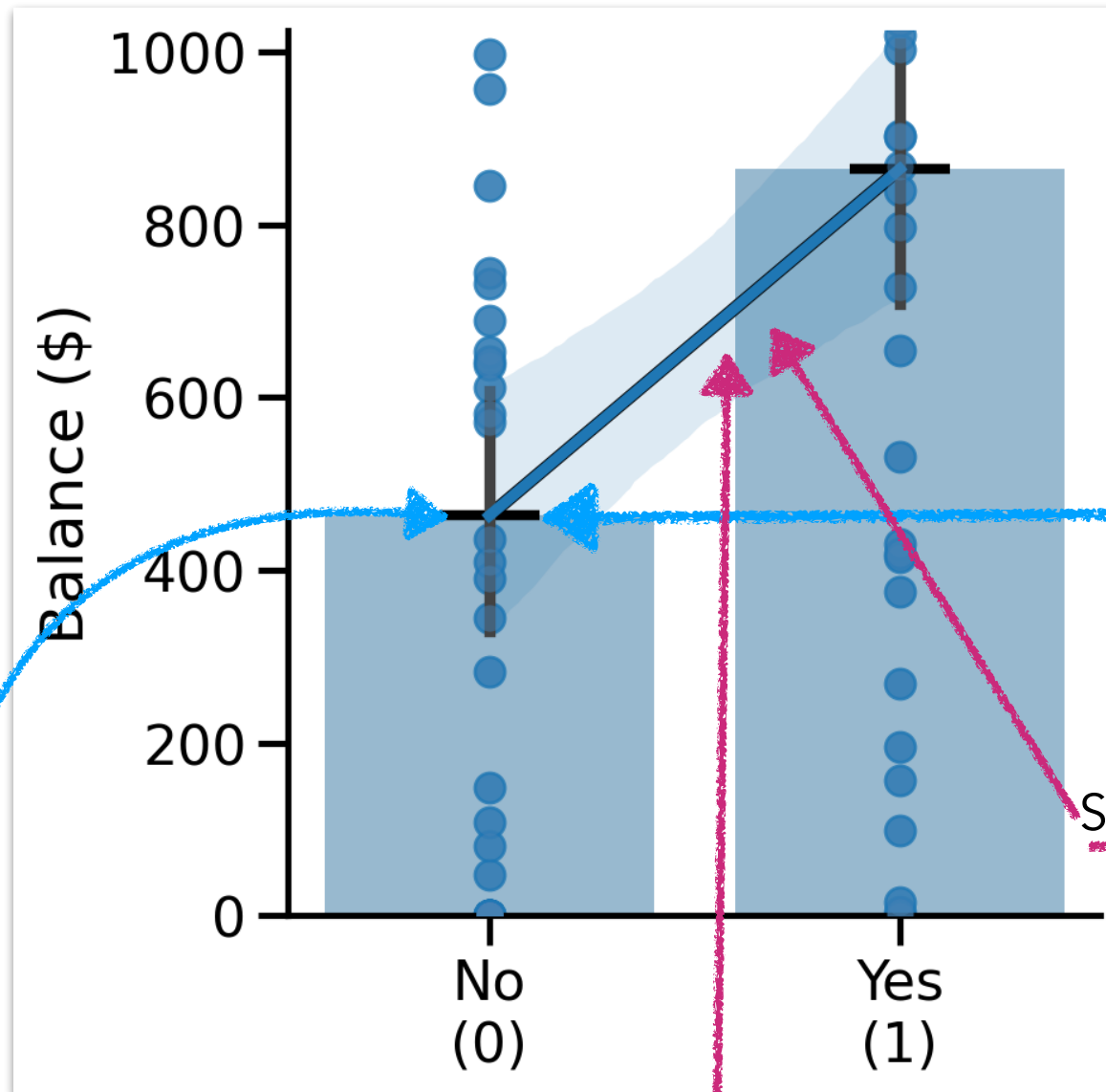


0: Student = No
1: Student = Yes

Student (Yes) - Student (No)

- **Reference level** is coded as 0, other level(s) coded as 1
- Intercept = reference-level (mean)
- **Slope(s)** = difference from reference-level mean

Independent Samples T-test



0: Student = No
1: Student = Yes

Student (Yes) - Student (No)

- **Reference level** is coded as 0, other level(s) coded as 1
- Intercept = reference-level (mean)
- **Slope(s)** = difference from reference-level mean

Categorical + continuous predictor

Dataset: Credit-Card Debt

index	income	limit	rating	cards	age	education	gender	student	married	ethnicity	balance
1	14.89	3606	283	2	34	11	Male	No	Yes	Caucasian	333
2	106.03	6645	483	3	82	15	Female	Yes	Yes	Asian	903
3	104.59	7075	514	4	71	11	Male	No	No	Asian	580
4	148.92	9504	681	3	36	11	Female	No	No	Asian	964
5	55.88	4897	357	2	68	16	Male	No	Yes	Caucasian	331
6	80.18	8047	569	4	77	10	Male	No	No	Caucasian	1151
7	21.00	3388	259	2	37	12	Female	No	No	African American	203
8	71.41	7114	512	2	87	9	Male	No	No	Asian	872
9	15.12	3300	266	5	66	13	Female	No	No	Caucasian	279
10	71.06	6819	491	3	41	19	Female	Yes	Yes	African American	1350

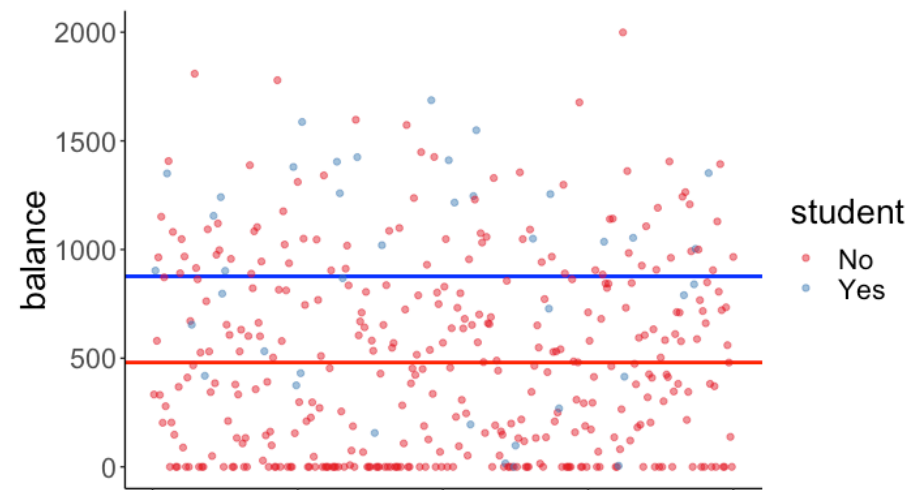
variable	description
income	in thousand dollars
limit	credit limit
rating	credit rating
cards	number of credit cards
age	in years
education	years of education
gender	male or female
student	student or not
married	married or not
ethnicity	African American, Asian, Caucasian
balance	average credit card debt in dollars

Do **students** have a different **credit card balance** from **non-students** when accounting for **income**?

Model C₁: 1 intercept; 1 slope

$$\text{balance}_i = \beta_0 + \beta_1 \text{student_dummy}_i$$

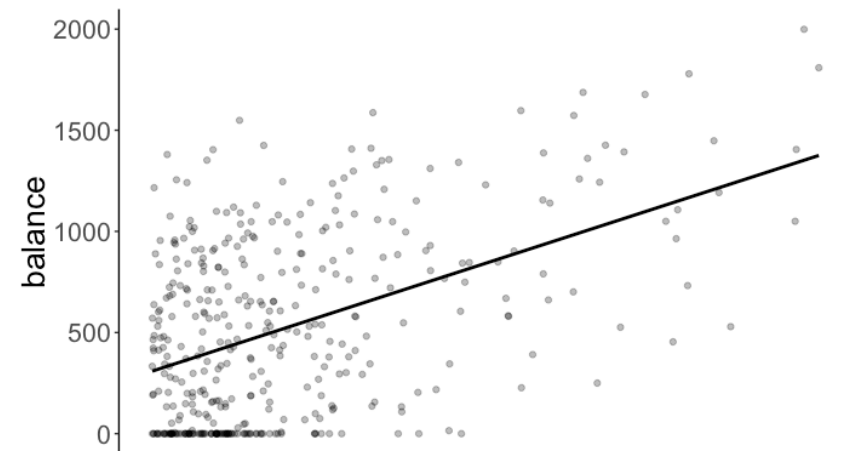
H₀₁: Students and non-students
have difference balances,
regardless of income



Model C₂: 1 intercept; 1 slope

$$\text{balance}_i = \beta_0 + \beta_1 \text{income}_i$$

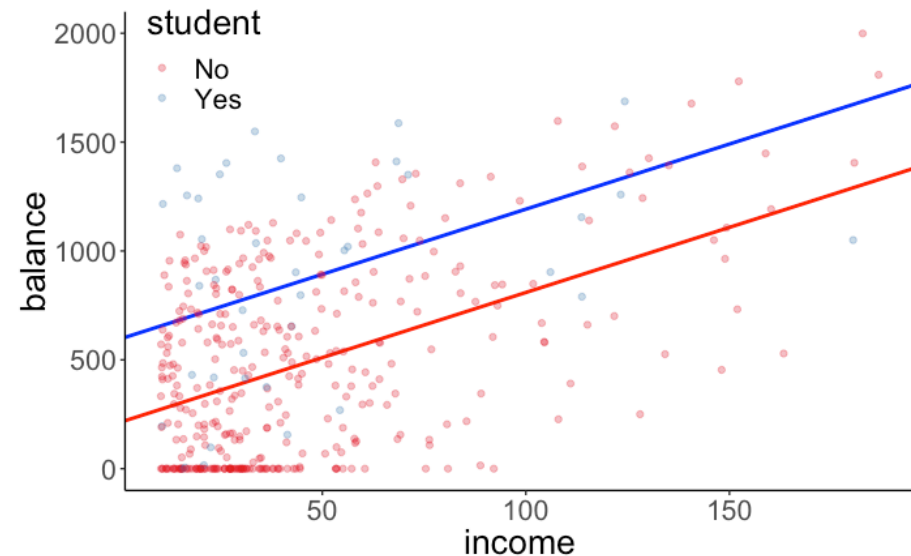
H₀₂: Students and non-students
have the same balances when
accounting for income



Model A: 2 intercepts; 1 slope

$$\text{balance}_i = \beta_0 + \beta_1 \text{student_dummy}_i + \beta_2 \text{income}_i$$

H₀₂: Students and non-students
have different balances when
accounting for income



Model A: 2 intercepts; 1 slope

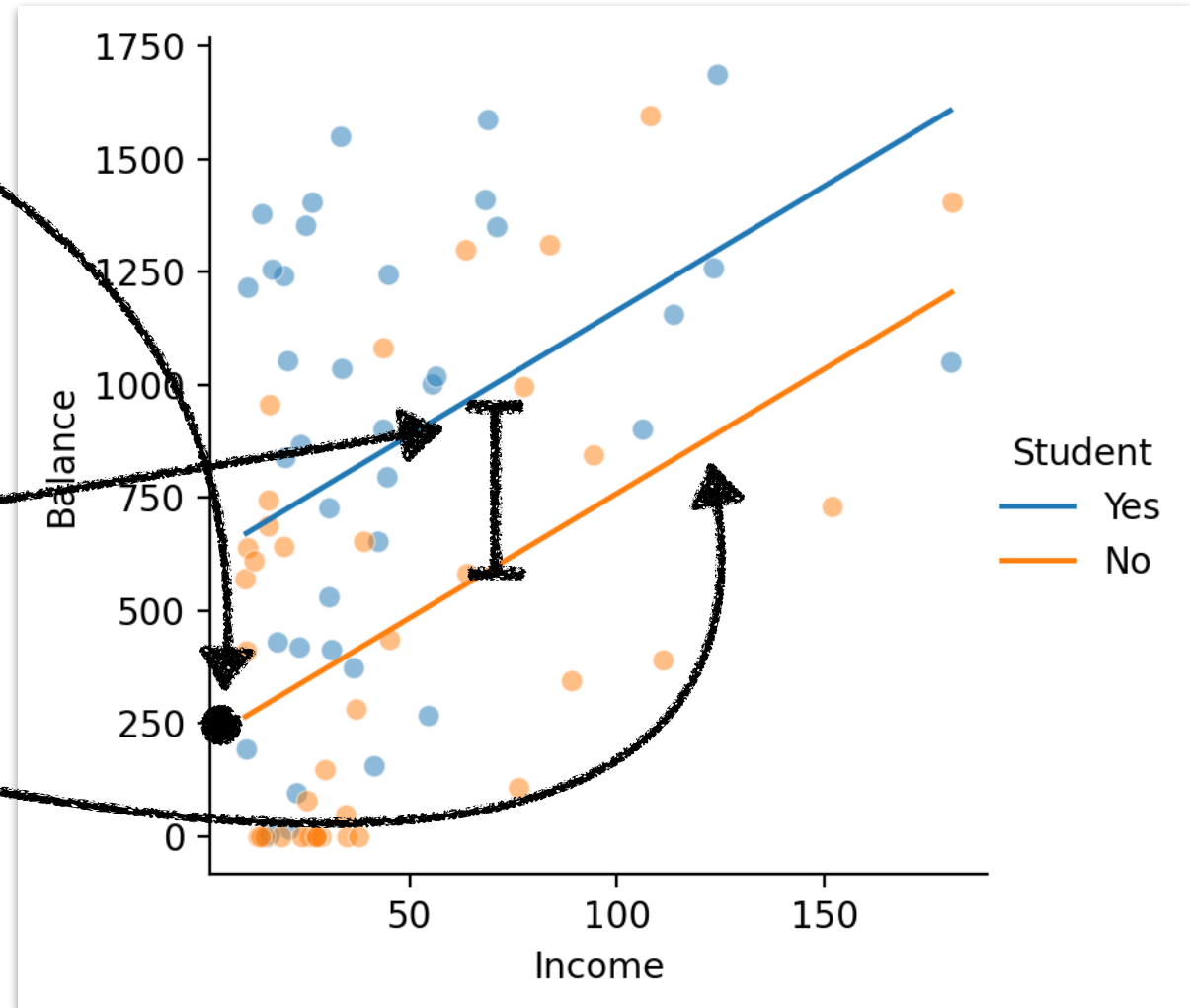
$$\text{balance}_i = \beta_0 + \beta_1 \text{student_dummy}_i + \beta_2 \text{income}_i$$

what do these represent?

$\hat{\beta}_0 = \text{Student} = 0 \text{ (No); Income} = 0$

$\hat{\beta}_1 = \text{Student (Yes - No); Income} = 0$

$\hat{\beta}_2 = \text{Income Slope; Student} = 0 \text{ (No)}$



Model A: 2 intercepts; 1 slope

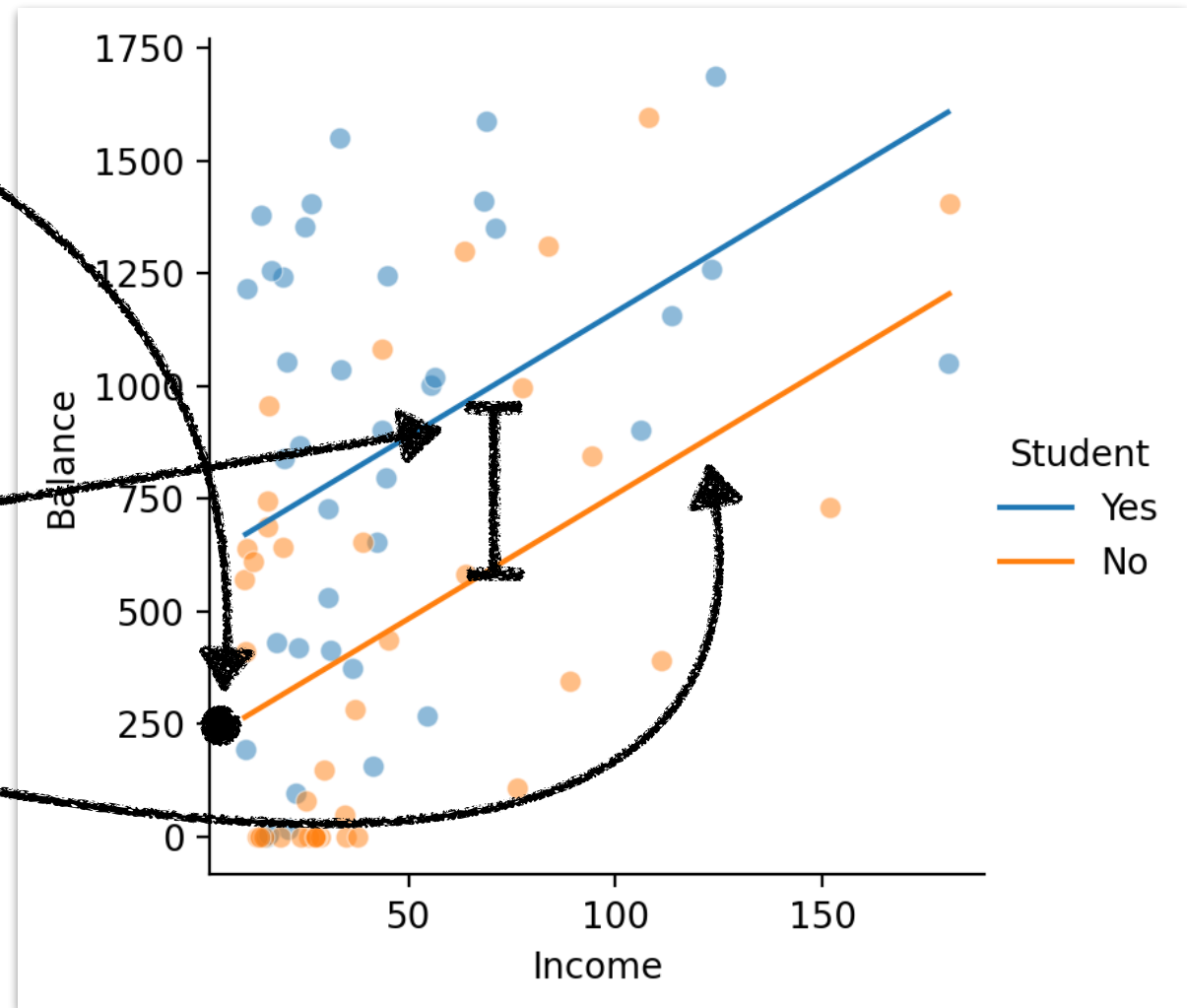
$$\text{balance}_i = \beta_0 + \beta_1 \text{student_dummy}_i + \beta_2 \text{income}_i$$

what do these represent?

$\hat{\beta}_0 = \text{Student} = 0 \text{ (No); Income} = 0$

$\hat{\beta}_1 = \text{Student (Yes - No); Income} = 0$

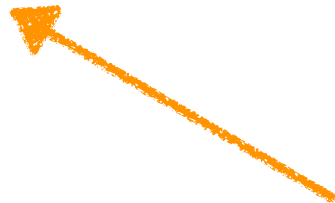
$\hat{\beta}_2 = \text{Income Slope; Student} = 0 \text{ (No)}$



Parameter Interpretation

- Whenever we have more than 1 predictor in a GLM
estimates assume other predictors = 0

$$\text{balance}_i = \beta_0 + \beta_1 \text{student_dummy}_i + \beta_2 \text{income}_i$$

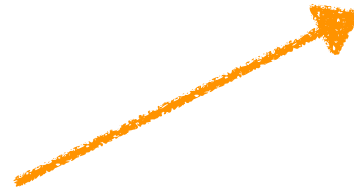


Slope of income
when student_dummy = 0

Parameter Interpretation

- Whenever we have more than 1 predictor in a GLM **estimates assume other predictors = 0**

$$\text{balance}_i = \beta_0 + \beta_1 \text{student_dummy}_i + \beta_2 \text{income}_i$$



Slope of student_dummy
when income = 0

- This is rarely ever what we want, unless 0 is a meaningful value for the predictor!
- But there's a simple solution: **centering predictors**
 - If we center a predictor(s) then estimates will assume **predictor(s) = centered value (typically mean)**

Model A: 2 intercepts; 1 slope centered income

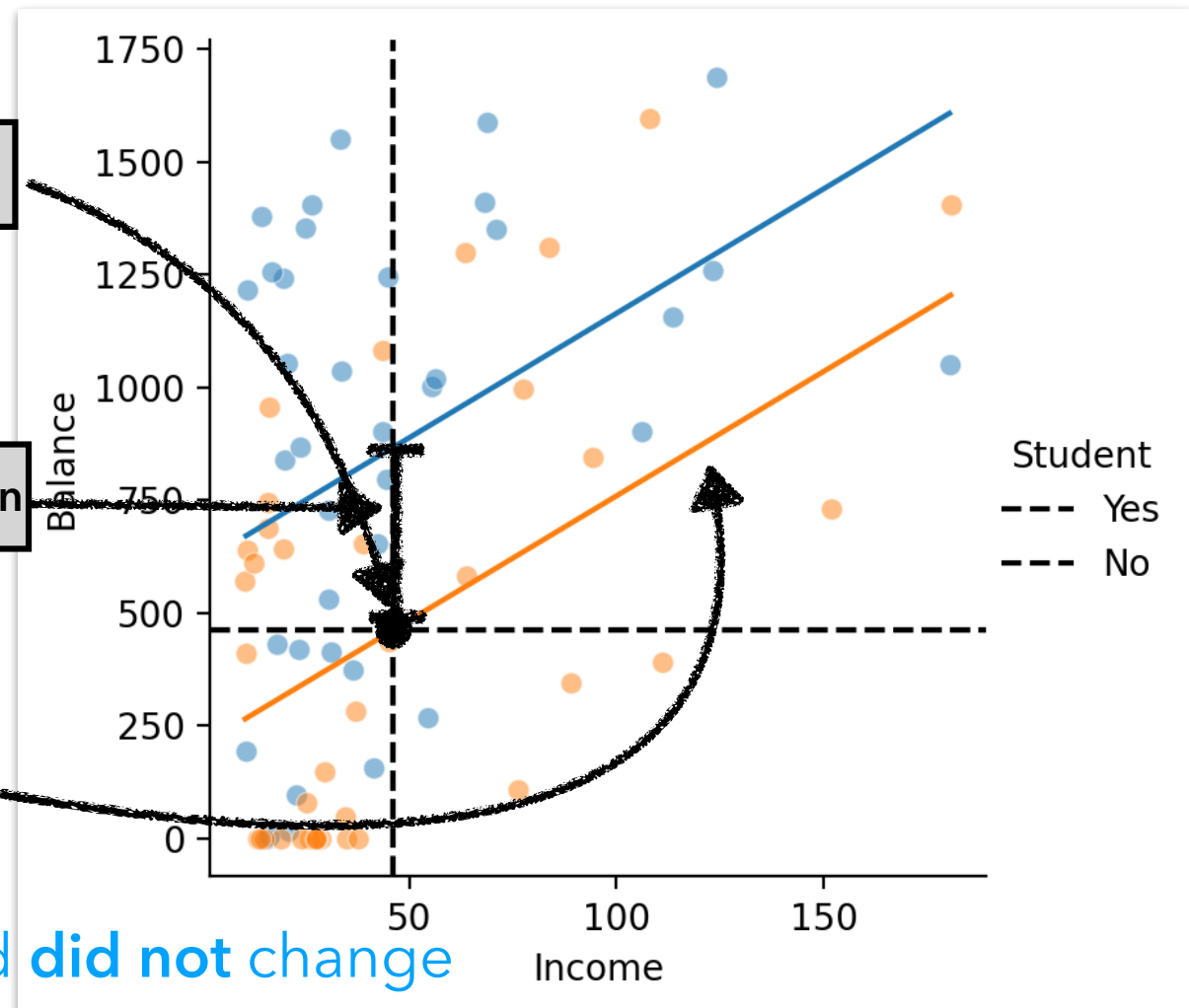
$$\text{balance}_i = \beta_0 + \beta_1 \text{student_dummy}_i + \beta_2 \text{income}_i$$

$\hat{\beta}_0 = \text{Student} = 0 \text{ (No)}; \text{Income} = \text{mean}$

$\hat{\beta}_1 = \text{Student (Yes - No)}; \text{Income} = \text{mean}$

$\hat{\beta}_2 = \text{Income Slope}; \text{Student} = 0 \text{ (No)}$

predictor we centered **did not change**



Parameter Interpretation

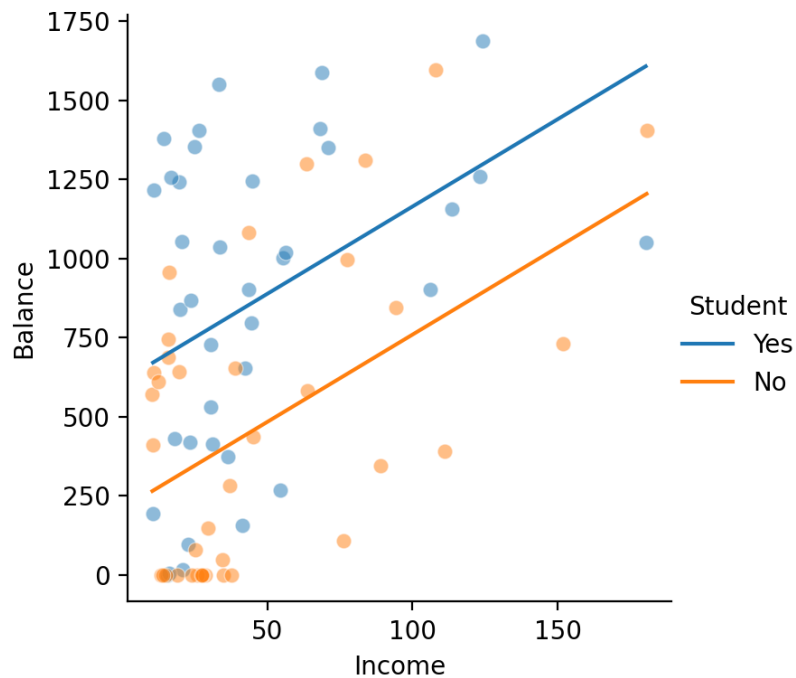
$$\text{balance}_i = \beta_0 + \beta_1 \text{student_dummy}_i + \beta_2 \text{income}_i$$

Original

$$\hat{\beta}_0 = \text{Student} = 0 \text{ (No); Income} = 0$$

$$\hat{\beta}_1 = \text{Student (Yes - No); Income} = 0$$

$$\hat{\beta}_2 = \text{Income Slope; Student} = 0 \text{ (No)}$$

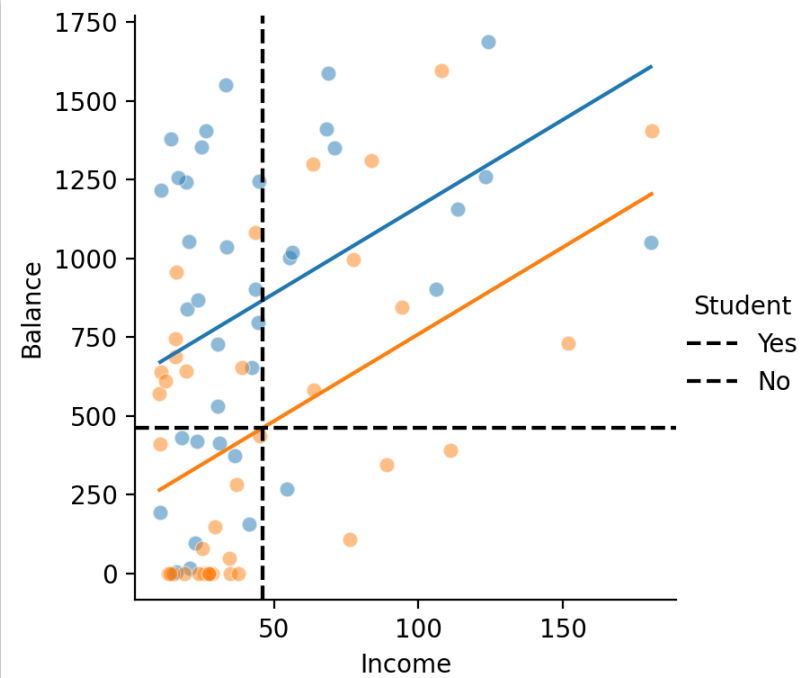


Centered

$$\hat{\beta}_0 = \text{Student} = 0 \text{ (No); Income} = \text{mean}$$

$$\hat{\beta}_1 = \text{Student (Yes - No); Income} = \text{mean}$$

$$\hat{\beta}_2 = \text{Income Slope; Student} = 0 \text{ (No)}$$



Parameter Interpretation Summary

$$\text{balance}_i = \beta_0 + \beta_1 \text{student_dummy}_i + \beta_2 \text{income}_i$$

- Whenever we have more than 1 predictor in a GLM **estimates assume other predictors = 0**
- This is rarely ever what we want, unless 0 is a meaningful value for the predictor!
- But there's a simple solution: **centering predictors**
 - If we center a predictor(s) then estimates will assume **predictor(s) = centered value (typically mean)**
- **Does NOT** change estimate of centered variable
- **Does NOT** change model predictions
- **Rule of Thumb:**
 - Whenever you have *any* categorical predictors, center your continuous predictors (it's almost always what you actually want to estimate)

For the rest of time (optional)

**Check out open datasets
for your final project**